PROJECT REPORT

Date	17 February 2026
Team ID	LTVIP2026TMIDS63806
Project Name	Heart Disease Analysis
Maximum Marks	

Heart Disease Analysis: A Data-Driven Exploration Using Tableau

1. INTRODUCTION

1.1 Background

Heart disease is one of the leading causes of death worldwide and remains a major public health concern. It is influenced by multiple factors including age, gender, lifestyle habits, diabetes, obesity, smoking, mental health, and other comorbid conditions.
Healthcare institutions and policymakers require accurate, data-driven insights to understand risk patterns, identify high-risk populations, and design effective prevention strategies. However, patient health data is often stored across spreadsheets, reports, and disconnected systems, making analysis time-consuming and complex.
Data visualization tools such as Tableau transform raw clinical datasets into interactive visual insights. These dashboards help stakeholders analyze demographic patterns, clinical indicators, and lifestyle factors contributing to heart disease.

1.2 Project Overview

Heart Disease Analysis is a Tableau-based data visualization project designed to analyze heart disease patterns across demographics, lifestyle factors, and health conditions.
This project focuses on:

- Gender-wise heart disease distribution
- Age-wise heart disease trends
- Race-wise analysis
- Impact of smoking and alcohol consumption
- Diabetes and stroke correlation
- BMI and age analysis
- Mental health and sleep patterns
- Comorbidity analysis (Asthma, Skin Cancer)
- Interactive storytelling using Tableau Story

The dashboard converts raw CSV datasets into interactive visuals that support informed healthcare decision-making.

1.3 Objectives

The main objectives of this project are:

- To design an interactive heart disease analytics dashboard
- To analyze demographic and clinical risk factors
- To identify high-risk age and population groups
- To study correlations between lifestyle habits and heart disease
- To present insights using story-based visualization
- To publish dashboards for web accessibility

2. IDEATION PHASE

2.1 Problem Statement

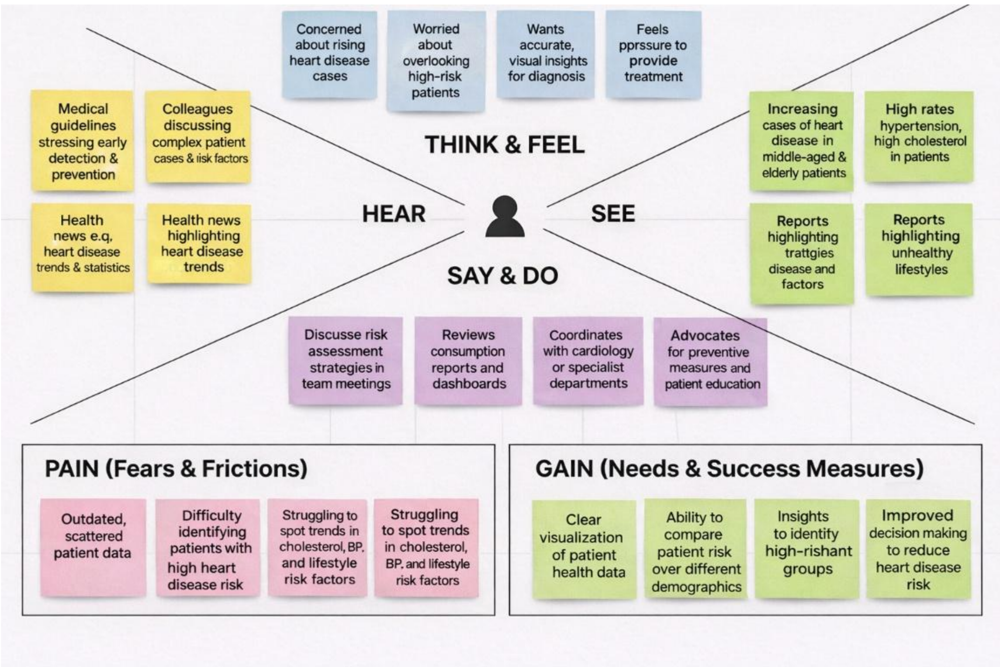
Healthcare professionals and public health authorities often lack a centralized, interactive platform to analyze heart disease risk patterns. Data stored in static reports limits the ability to identify correlations, monitor risk factors, and support preventive healthcare planning.

2.2 Target Users

The primary users of this dashboard include:

- Cardiologists and general physicians
- Hospital administrators
- Public health researchers
- Healthcare policymakers
- Academic researchers
- Health analytics professionals

2.3 Empathy Map



THINK & FEEL

- Concerned about rising heart disease cases
- Worried about delayed diagnosis
- Wants accurate risk assessment tools
- Feels pressure to improve patient outcomes
- Aims to reduce preventable deaths

HEAR

- Patients concerned about lifestyle risks
- Government promoting preventive healthcare
- Media discussing rising cardiovascular diseases
- Medical experts emphasizing early screening

SEE

- Increasing cases in middle-aged populations
- Higher risk among smokers and diabetic patients
- Obesity and sedentary lifestyle trends
- Mental health issues linked to cardiovascular risk

SAY & DO

- Reviews patient records and reports
- Discusses preventive strategies in meetings
- Encourages lifestyle modification programs
- Uses dashboards for data-driven insights

PAIN (Fears & Frictions)

- Difficulty identifying high-risk patients early
- Scattered patient data across systems
- Limited real-time analytical tools
- Unclear correlations between health indicators
- Delayed decision-making

GAIN (Needs & Success Measures)

- Clear visualization of health risk patterns
- Ability to compare age groups and genders
- Insights into lifestyle and clinical correlations
- Improved diagnosis support
- Data-driven healthcare planning

3. REQUIREMENT ANALYSIS

3.1 Functional Requirements

- Interactive filters (Age, Gender, Race, BMI, Diabetes, Smoking)
- Gender-wise heart disease visualization
- Age-based trend charts
- Diabetes vs Stroke analysis
- Mental health and sleep analysis
- Risk factor comparison visuals
- Story-based dashboard navigation
- KPI summary indicators (Total Patients, High-Risk %, Avg BMI)

3.2 Non-Functional Requirements

- Dashboard loading time under 5 seconds
- Responsive layout for presentations
- Accurate KPI calculations
- Secure publishing via Tableau Public
- Scalability for additional medical datasets

3.3 Data Requirements

The dashboard uses electricity consumption datasets containing:

- Age
- Gender
- Race
- BMI
- Smoking status
- Alcohol consumption
- Diabetes status
- Stroke history
- General health status
- Mental health indicators
- Sleep time

Total Records: 4000+ entries

4. SYSTEM DESIGN

4.1 Data Flow Diagram

4.2 Solution Architecture

User Interface Layer:

- Tableau Dashboard (Web-based)
- Interactive filters and KPI cards

Application Layer:

- Tableau Calculated Fields
- Data filtering and aggregation logic

Data Layer:

- CSV datasets
- Optional cloud storage

Deployment Layer:

- Tableau Public
- Embedded via HTML iframe

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning Table

Phase	Task	Time (Days)
Phase -1	Data Collection	1
Phase -2	Data Cleaning & Preprocessing	2
Phase -3	Visualizations	2
Phase -4	Dashboard & Stories	2
Phase - 5	Performance Testing	1
Phase-6	Web Integration & Publishing	1
Phase - 7	Report & documentation	3

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

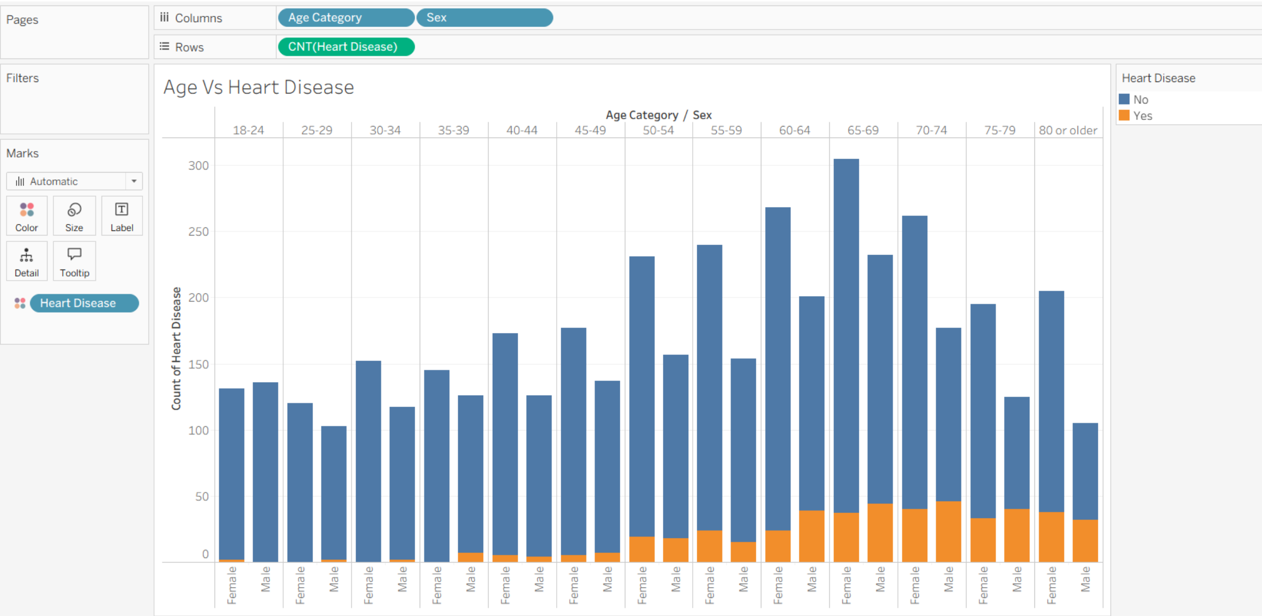
- Data Loaded: 4000+ records
- Filters Used: Age, Gender, Race, BMI, Diabetes, Smoking
- Calculated Fields: Risk %, BMI Average, Age Distribution
- Visualizations: 15+
- Result: Smooth performance with response time under 3–5 seconds

7. RESULTS

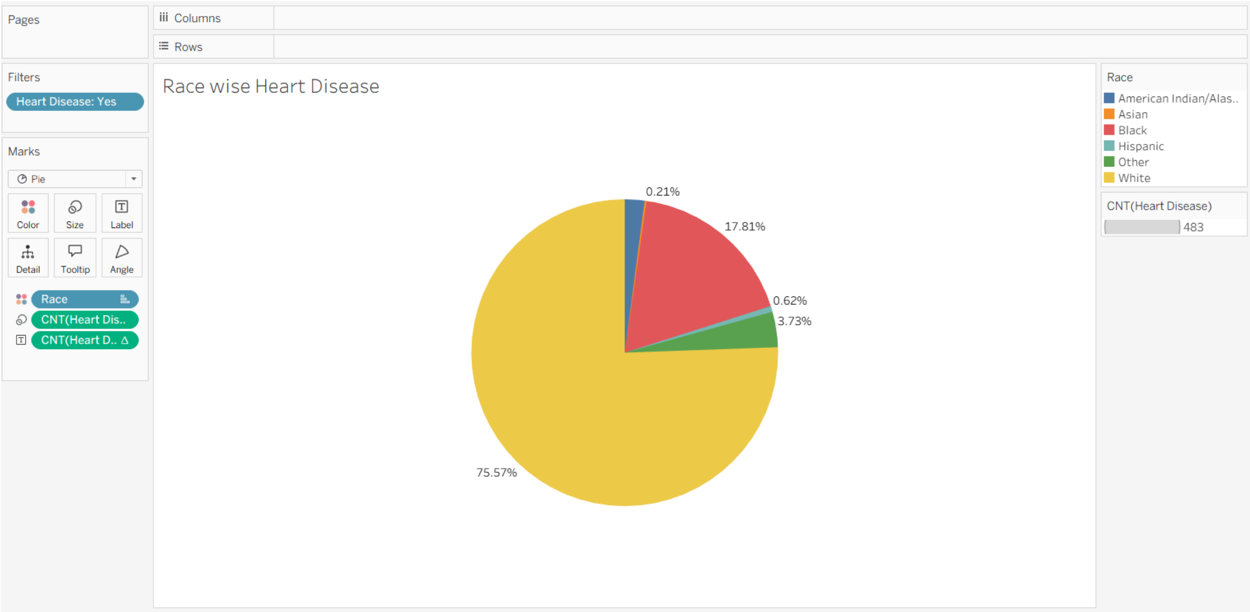
7.1 Output Screenshots

➢ **Gender Vs Heart Disease**

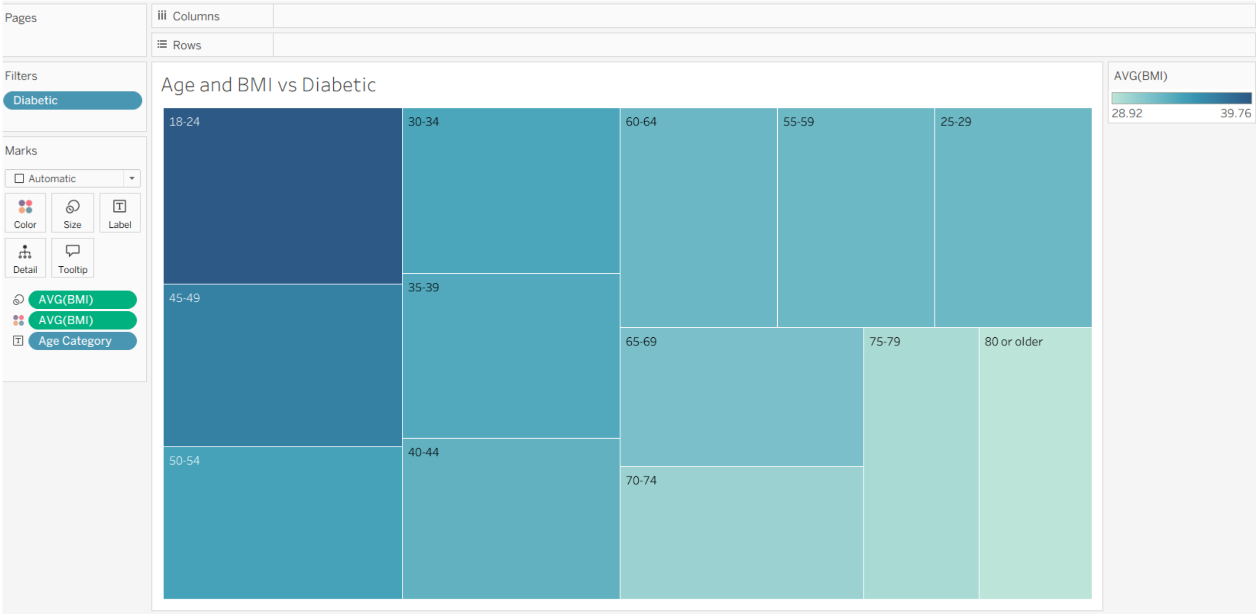
➢ **Age Vs Heart Disease**



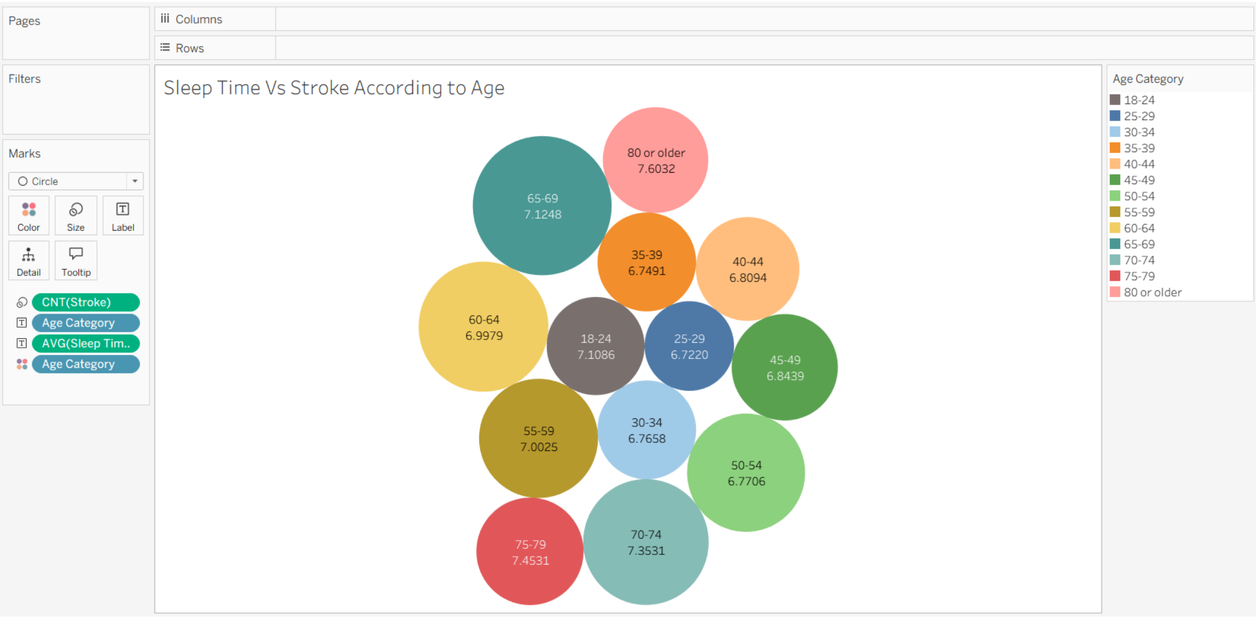
➢ Race wise Heart Disease



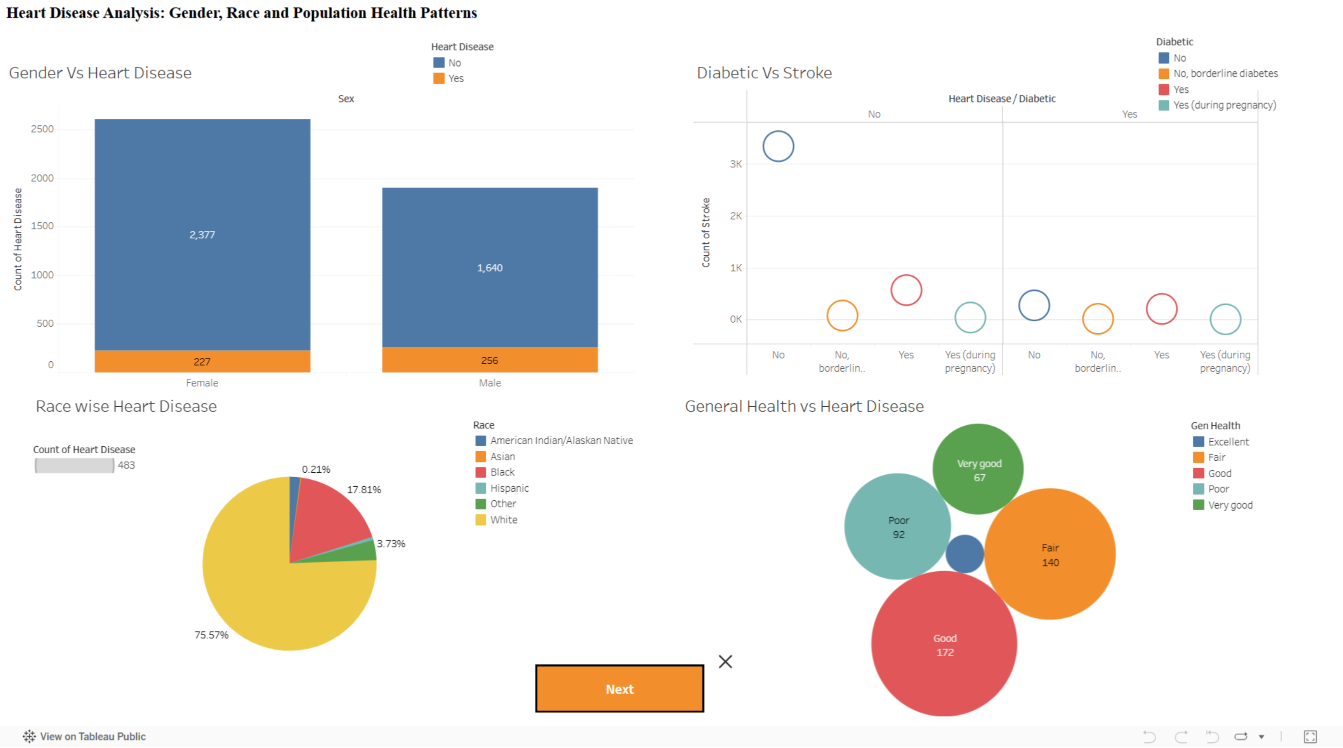
➢ Age and BMI Vs Diabetic



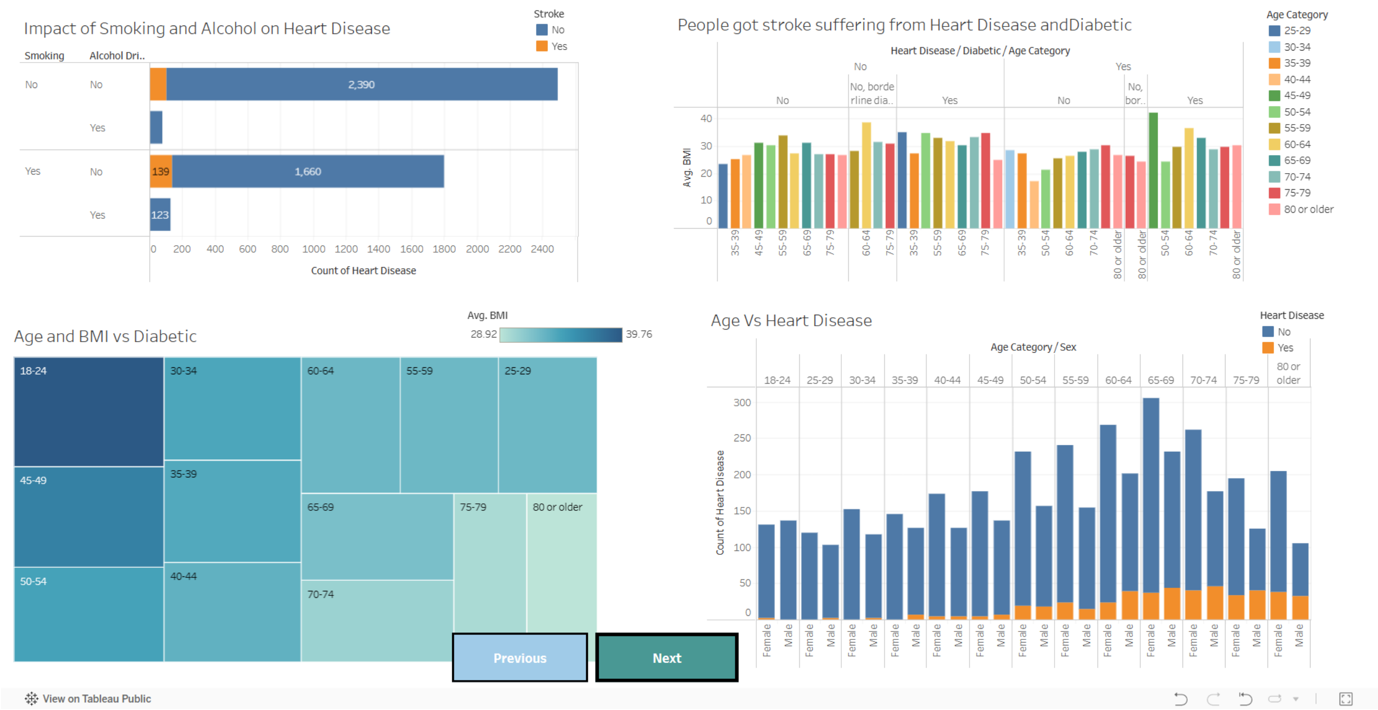
➢ Sleep Time Vs Stroke According to Age



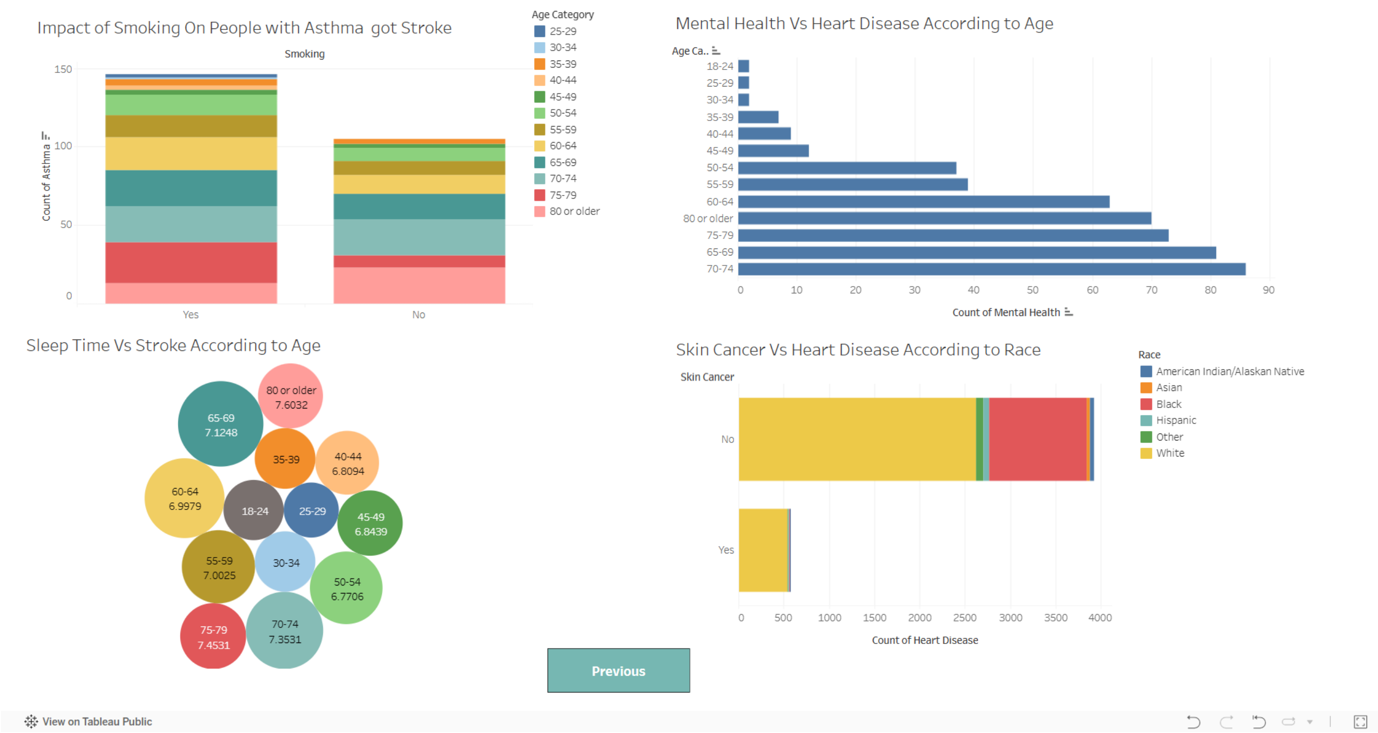
➢ Dashboard:



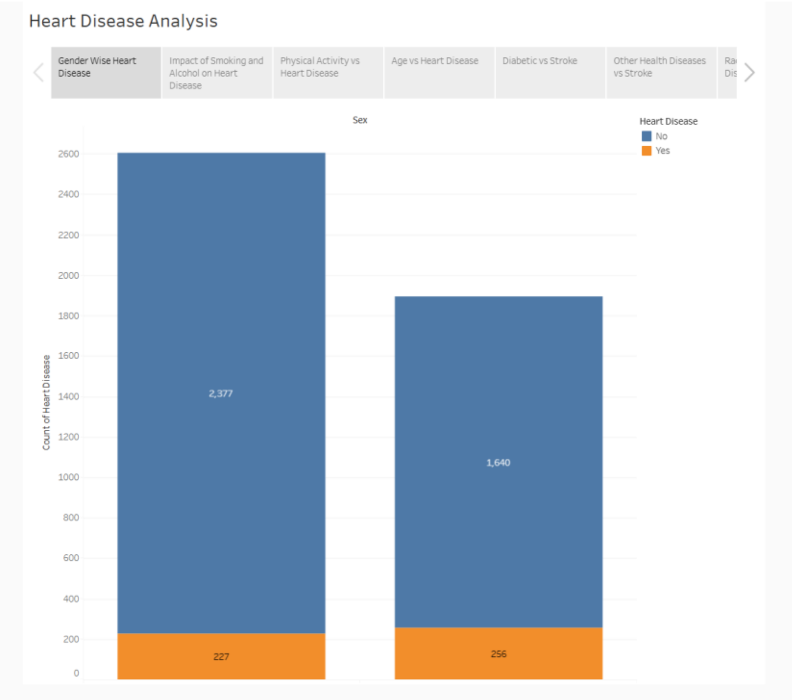
Heart Disease Analysis: Impact of Smoking, Alcohol and BMI

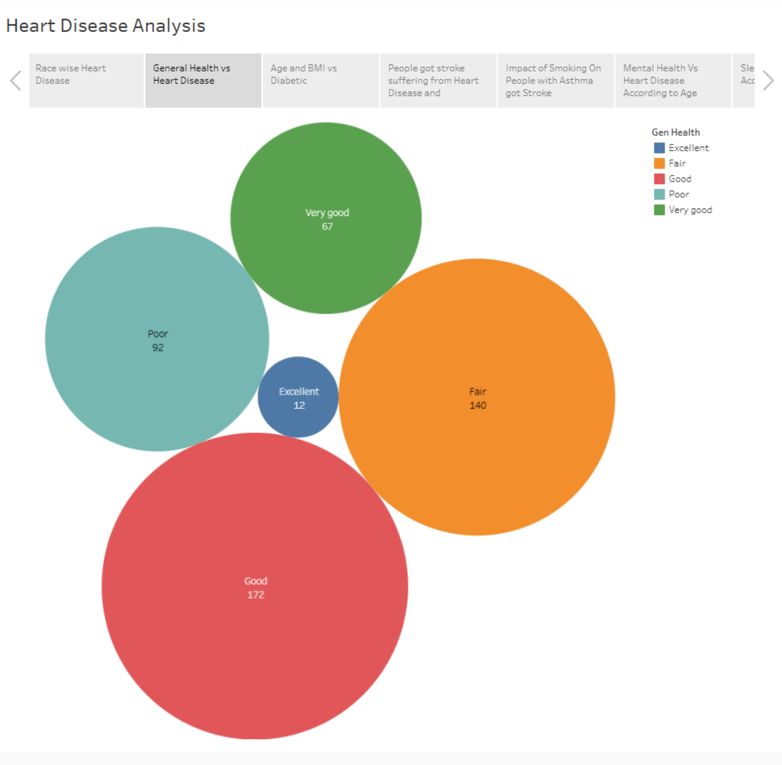


Heart Disease Analysis: Mental Health, Sleep and Secondary Conditions



➢ Story


Heart Disease Analysis
Gender Wise Heart Disease
Impact of Smoking and Alcohol on Heart Disease
Physical Activity vs Heart Disease
Age vs Heart Disease
Diabetic vs Stroke
Other Health Diseases vs Stroke
Sex
Heart Disease
No
Yes
Count of Heart Disease
2600
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
2,377
227
1,640
256


Heart Disease Analysis
Race wise Heart Disease
General Health vs Heart Disease
Age and BMI vs Diabetic
People got stroke suffering from Heart Disease and
Impact of Smoking On People with Asthma got Stroke
Mental Health Vs Heart Disease According to Age
Gen Health
Excellent
Fair
Good
Poor
Very good
Very good
67
Poor
92
Excellent
12
Fair
140
Good
172

8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy interpretation of health risk patterns
- Centralized heart disease insights
- Interactive and story-driven presentation
- Supports preventive healthcare planning
- Identifies high-risk population groups

Disadvantages

- Requires Tableau knowledge
- Uses static datasets (no live hospital integration)
- Limited scalability without cloud or EHR integration

9. CONCLUSION

This project demonstrates how Tableau can transform raw clinical datasets into meaningful visual insights. The Heart Disease Analysis dashboard simplifies complex health data into interactive analytics, enabling healthcare professionals to identify risk patterns, analyze correlations, and support preventive healthcare strategies.

10. FUTURE SCOPE

- Integrate real-time hospital and EHR data
- Implement predictive risk modeling using Machine Learning
- Add heart disease risk scoring system
- Develop mobile-friendly healthcare dashboards
- Expand analysis to multi-hospital datasets
- Integrate wearable health monitoring data

11. APPENDIX

Source Code(Web integration)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Heart Disease Analysis</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap"
rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    scroll-behavior:smooth;
}
```

```
/* HERO */
.hero-section{
   background:linear-gradient(rgba(0,0,0,.65),rgba(0,0,0,.65)),
   url('https://cloudfront-us-east-
1.images.arcpublishing.com/infobae/XIX6B6CPDVEQ7BWFSYNFY6HVLY.png');
   background-size:cover;
   background-position:center;
   height:100vh;
   color:#fff;
   display:flex;
   align-items:center;
   text-align:center;
}

.hero-content h1{
   font-size:3rem;
   font-weight:700;
}

.btn-custom{
   background:#e84118;
   color:#fff;
   padding:12px 30px;
   border-radius:30px;
   font-weight:600;
   transition:.3s;
}

.btn-custom:hover{
   background:#c23616;
   transform:translateY(-2px);
}

section{padding:80px 0;}

.section-title{
   text-align:center;
   margin-bottom:50px;
}

.section-title h2{
   font-weight:600;
   position:relative;
}

.section-title h2::after{
   content:'';
   width:50px;
   height:3px;
```

```
    background:#e84118;
    position:absolute;
    bottom:-8px;
    left:50%;
    transform:translateX(-50%);
}

.tableau-container{
    background:#fff;
    padding:20px;
    border-radius:15px;
    box-shadow:0 10px 30px rgba(0,0,0,0.1);
}

.contact-card{
    text-align:center;
    padding:30px;
    border:1px solid #eee;
    border-radius:10px;
    transition:.3s;
}

.contact-card:hover{
    box-shadow:0 10px 20px rgba(0,0,0,0.05);
    transform:translateY(-5px);
}

.contact-icon{
    font-size:2rem;
    color:#e84118;
}

footer{
    background:#2d3436;
    color:#fff;
    padding:20px;
    text-align:center;
}
</style>
</head>

<body data-bs-spy="scroll" data-bs-target="#navbar">

<!-- NAVBAR -->
<nav id="navbar" class="navbar navbar-expand-lg navbar-dark bg-dark fixed-top">
<div class="container">
<a class="navbar-brand fw-bold" href="#">Heart Disease Analysis</a>

<button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#nav">
```

```html
<span class="navbar-toggler-icon"></span>
</button>

<div class="collapse navbar-collapse" id="nav">
<ul class="navbar-nav ms-auto">
<li class="nav-item"><a class="nav-link active" href="#home">Home</a></li>
<li class="nav-item"><a class="nav-link" href="#about">About</a></li>
<li class="nav-item"><a class="nav-link" href="#dashboard">Dashboard</a></li>
<li class="nav-item"><a class="nav-link" href="#story">Story</a></li>
<li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>
</ul>
</div>
</div>
</nav>

<!-- HERO -->
<section id="home" class="hero-section">
<div class="container hero-content">
<h1>Heart Disease Analysis</h1>
<p class="lead">Data-driven insights to understand cardiovascular risk factors and support early
detection.</p>
<a href="#dashboard" class="btn btn-custom">View Dashboard</a>
</div>
</section>

<!-- ABOUT -->
<section id="about">
<div class="container">
<div class="section-title">
<h2>About the Project</h2>
</div>

<div class="row align-items-center">
<div class="col-md-6">
<img src="https://cloudfront-us-east-
1.images.arcpublishing.com/infobae/XIX6B6CPDVEQ7BWFSYNFY6HVLY.png"
class="img-fluid rounded shadow" alt="Heart health analysis">
</div>

<div class="col-md-6 mt-4 mt-md-0">
<h4>Why Heart Disease Analysis?</h4>
<p class="text-muted">
Heart disease remains one of the leading causes of death worldwide. This project uses data visualization
to identify risk patterns, monitor health indicators, and support preventive care decisions.
</p>

<ul class="list-unstyled">
<li class="mb-2">📊 <strong>Dashboard:</strong> Overview of patient
demographics & risk factors.</li>
```

```
<li class="mb-2">📖 <strong>Story:</strong> Guided insights into prevention & trends.</li>
<li>❤️ <strong>Healthcare Insights:</strong> Promoting awareness & data-driven decisions.</li>
</ul>
</div>
</div>
</div>
</section>

<!-- DASHBOARD -->
<section id="dashboard" class="bg-light">
<div class="container">
<div class="section-title">
<h2>Dashboard</h2>
<p>Interactive visualization of heart disease risk indicators</p>
</div>

<div class="tableau-container">
<div class='tableauPlaceholder' id='vizDashboard'></div>
</div>
</div>
</section>

<!-- STORY -->
<section id="story">
<div class="container">
<div class="section-title">
<h2>Story Insights</h2>
<p>Understanding patterns and prevention</p>
</div>

<div class="tableau-container">
<div class='tableauPlaceholder' id='vizStory'></div>
</div>
</div>
</section>

<!-- CONTACT -->
<section id="contact" class="bg-light">
<div class="container">
<div class="section-title">
<h2>Contact Team</h2>
<p>Reach out for collaboration & insights</p>
</div>

<div class="row justify-content-center">
<div class="col-md-4">
<div class="contact-card bg-white">
<div class="contact-icon">📊</div>
<h5>Shaik Farukh Ahamed</h5>
```

```html
<p class="text-muted">Data Analyst</p>
<p><strong>Email:</strong><br>
<a href="mailto:farukhahamed456@gmail.com">farukhahamed456@gmail.com</a>
</p>
</div>
</div>
</div>

</div>
</section>

<footer>
<p>&copy; 2026 Heart Disease Analysis Project</p>
</footer>

<!-- Tableau Embeds -->
<script src="https://public.tableau.com/javascripts/api/viz_v1.js"></script>
<script>
function loadViz(id, url){
   const container = document.getElementById(id);
   const viz = document.createElement("iframe");
   viz.src = url;
   viz.style.width="100%";
   viz.style.height="800px";
   viz.style.border="none";
   container.appendChild(viz);
}

loadViz("vizDashboard",
"https://public.tableau.com/views/HeartDiseaseanalysis_17720896987490/Dashboard1?
:showVizHome=no");

loadViz("vizStory",
"https://public.tableau.com/views/HeartDiseaseanalysisstory_17720901197950/HeartDiseaseAnalysis?
:showVizHome=no");
</script>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Data set Link:-
CSV file used in Tableau

https://drive.google.com/file/d/1Q6L-eiRaoQV5vhXY_ghNWyf9uPmkoQjD/view?usp=sharing

Tableau Publish:-

I. Dashboard:
https://public.tableau.com/views/HeartDiseaseanalysis_17720896987490/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

II. Story
https://public.tableau.com/views/HeartDiseaseanalysisstory_17720901197950/HeartDiseaseAnalysis?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link